ACTIVITY 2: CYBER RANGE

Cyber Range as a Service (CRaaS)

Topic: Ransomware Attack vs File Recovery Defense

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1. Introduction

Ransomware has emerged as one of the most destructive and financially devastating categories of cyber threats in modern times. From crippling hospital networks to disrupting critical government infrastructure, ransomware attacks have demonstrated that no organisation is immune. Understanding how such attacks are orchestrated — and how defenders can respond effectively — is a cornerstone of modern cybersecurity education.

This activity presents a GUI-based Python simulation of a Ransomware Attack vs File Recovery Defense scenario, developed within the Cyber Range as a Service (CRaaS) framework. The simulation provides a safe, fully isolated, and interactive environment where students can:

- Execute a simulated end-to-end ransomware kill chain against a sandbox file system
- Observe file encryption, ransom note deployment, and attacker communication in real time
- Apply multiple incident response and recovery strategies as a defender
- Analyse Indicators of Compromise (IoCs), incident timelines, and defence scoring

1.1 Motivation

According to Cybersecurity Ventures, global ransomware damage costs exceeded USD 20 billion in 2021 and continue to rise year over year. High-profile attacks such as WannaCry, NotPetya, and the Colonial Pipeline incident have underscored how ransomware can paralyse entire industries. Digital forensics practitioners must be equipped not only to investigate such incidents post-mortem but to understand the attacker's methodology at a technical level.

This simulation bridges the gap between theoretical knowledge and practical, hands-on experience — making it an ideal activity for MSc DFIS students studying incident response and cyber threat intelligence.

2. Conceptual Background

2.1 Cyber Range as a Service (CRaaS)

A Cyber Range is a controlled, virtualised environment designed to simulate real-world cyberattack scenarios. Cyber Range as a Service (CRaaS) extends this concept by delivering such environments on demand via cloud infrastructure, removing the need for dedicated physical labs and making high-quality security training accessible to students and practitioners anywhere.

The key principles of a CRaaS environment are:

- Isolation: All simulated activity is confined to a sandboxed environment, with zero risk to real systems or files
- Realism: Simulations replicate actual attacker techniques, tools, and procedures (TTPs)
- Measurability: All events are logged, scored, and time-stamped for forensic analysis and learning feedback
- Scalability: Scenarios can be spun up and torn down rapidly for multiple concurrent users

2.2 Ransomware — Overview

Ransomware is a category of malicious software that encrypts a victim's files and demands payment (typically in cryptocurrency) in exchange for the decryption key. Modern ransomware variants follow a well-documented attack lifecycle, often referred to as the Kill Chain model.

The five key stages of a ransomware kill chain are:

#	Stage	Description
1	Initial Access	Entry via phishing email, malicious attachment, RDP exploitation, or drive-by download
2	Key Generation	A unique AES symmetric encryption key is generated and optionally transmitted to the C2 server
3	File Enumeration	The malware recursively scans the file system for target file types (documents, images, databases)
4	Encryption	Each identified file is encrypted and renamed with a .locked or custom extension, rendering it inaccessible
5	Ransom Note Delivery	A text or HTML ransom note is dropped containing payment instructions, wallet address, and deadline

2.3 Incident Response Strategies

Incident response to a ransomware attack follows the NIST Cybersecurity Framework phases: Identify, Protect, Detect, Respond, and Recover. The three primary recovery strategies modelled in this simulation are:

- Backup Restoration (Strategy A): Restoring files from a pre-attack clean backup is universally regarded as the gold standard response. It requires no ransom payment and results in full data recovery, provided backups are current and stored offline or in an isolated environment.
- Decryption Using Captured Key (Strategy B): In some scenarios — particularly during law enforcement takedowns of ransomware operations — decryption keys are recovered. This strategy simulates that scenario, allowing affected files to be restored cryptographically.
- Network Isolation / Containment (Strategy C): When recovery is not immediately possible, network isolation — severing the affected system from all network connectivity — is employed to prevent lateral movement and stop the exfiltration of the encryption key to the attacker's Command and Control (C2) server.

2.4 Indicators of Compromise (IoCs)

Indicators of Compromise are forensic artefacts that indicate a system has been compromised. In the context of ransomware, key IoCs include: rapid mass file renaming events, creation of ransom note files, sudden spikes in CPU or disk I/O, unexpected outbound network connections to C2 infrastructure, and the appearance of new file extensions not previously present on the system.

This simulation tracks and logs all such IoCs in real time, providing students with a forensically relevant event record that mirrors what a digital forensics analyst would encounter during a post-incident investigation.

3. Implementation

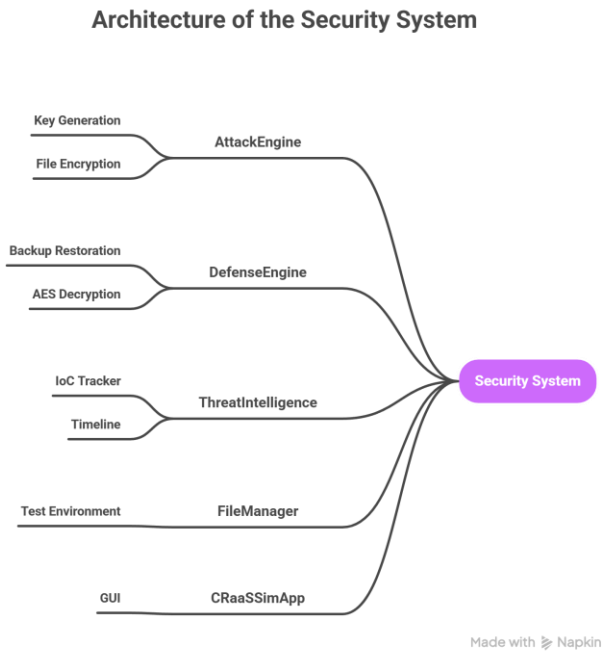
3.1 Technology Stack

- Language: Python 3.x
- GUI Framework: Tkinter (Python standard library)
- Encryption: cryptography library — Fernet (AES-128-CBC with HMAC-SHA256)
- File Operations: os, shutil, pathlib (Python standard library)
- Logging: logging module with rotating file handler
- Reporting: json module for structured incident report generation
- Concurrency: threading for non-blocking GUI during attack/recovery operations

3.2 System Architecture

The simulation follows a modular five-component architecture. Each module is independently responsible for a specific phase of the attack-defense lifecycle, ensuring clean separation of concerns:

Module	Team	Responsibilities
AttackEngine	Red Team	Key generation, file enumeration, AES encryption, .locked renaming, ransom note deployment
DefenseEngine	Blue Team	Pre-attack backup creation, backup restoration, key-based decryption, network isolation simulation
ThreatIntelligence	Analyst	IoC collection, incident timeline, TTD calculation, defence scoring, JSON report generation
FileManager	Support	Sandbox directory creation with 15 realistic sample files, safe cleanup post-simulation
CRaaSSimApp	Interface	Tkinter GUI integrating all modules; real-time logs, progress indicators, and interactive controls



3.3 Encryption Implementation

The simulation uses the Fernet symmetric encryption scheme from the Python cryptography library. Fernet implements AES-128 in CBC mode with a SHA-256 HMAC for message authentication, making it a production-grade, authenticated encryption scheme that closely models real-world ransomware behaviour.

3.3.1 Key Generation

```
from cryptography.fernet import Fernet
self.encryption_key = Fernet.generate_key()
self.cipher         = Fernet(self.encryption_key)
```

3.3.2 File Encryption

```
with open(filepath, 'rb') as f:
    plaintext  = f.read()
ciphertext = self.cipher.encrypt(plaintext)
locked_path = filepath.with_suffix('.locked')
with open(locked_path, 'wb') as f:
    f.write(ciphertext)
filepath.unlink()   # delete original
```

3.3.3 Decryption (Recovery)

```
with open(locked_file, 'rb') as f:
    ciphertext = f.read()
plaintext = self.cipher.decrypt(ciphertext)
```

```
with open(original_path, 'wb') as f:
    f.write(plaintext)
```

3.4 Output Files Generated

File / Directory	Purpose
craas_test_files/	Sandbox directory containing 15 sample files targeted by the attack
craas_test_files_backup/	Backup copy created by DefenseEngine before attack begins
RANSOM_NOTE.txt	Simulated ransom demand dropped by AttackEngine post-encryption
incident_report.json	Structured forensic report generated by ThreatIntelligence module
craas_sim.log	Chronological event log with timestamps for all simulation activities

4. Working Process

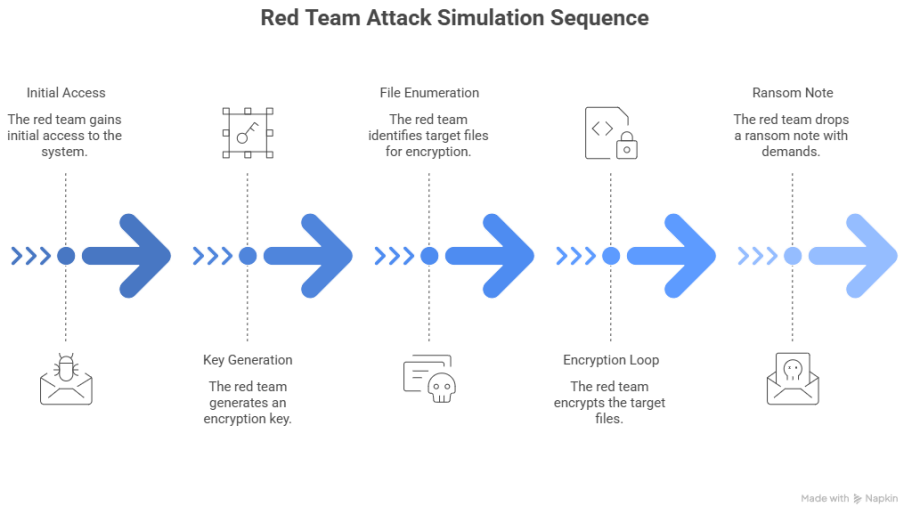
4.1 Pre-Simulation Setup

Before the attack begins, the FileManager module initialises the sandbox environment by creating 15 realistic sample files within the craas_test_files/ directory. These files simulate a typical user's documents, including text files, CSV data, configuration files, and log files. Simultaneously, the DefenseEngine creates a complete backup of this directory in craas_test_files_backup/ to enable Strategy A recovery later.

4.2 Attack Phase — Red Team Kill Chain

When the user initiates the attack via the GUI, the AttackEngine executes the following five-stage kill chain:

1. Initial Access: The simulation logs a mock phishing or payload delivery event as the attack entry point, along with a timestamp recorded by ThreatIntelligence
2. Key Generation: Fernet.generate_key() produces a 256-bit URL-safe base64-encoded encryption key. This key is stored in memory (simulating exfiltration to a C2 server in real ransomware) and displayed in the GUI for use in recovery Strategy B
3. File Enumeration: The AttackEngine scans all files in craas_test_files/ using pathlib.Path.rglob(), filtering for non-.locked files and logging each discovered file as an IoC
4. Encryption: Each file is read, encrypted with the Fernet cipher, written with a .locked extension, and the original file is deleted. The GUI updates a progress indicator after each file is processed
5. Ransom Note Deployment: RANSOM_NOTE.txt is created containing the simulated attacker's payment demand, a mock wallet address, and a 72-hour deadline. This event is logged as a critical IoC



4.3 Defense Phase — Blue Team Response

After the attack completes, the user can select from three recovery strategies through the GUI:

Strategy A — Backup Restoration

The DefenseEngine copies all original files from craas_test_files_backup/ back to craas_test_files/, overwriting the .locked files. This is the fastest and most complete recovery method, representing best-practice cybersecurity posture. The Defence Score is maximised when this strategy is applied promptly after detection.

Strategy B — Decryption Using Captured Key

Using the encryption key generated during Stage 2 of the attack, the DefenseEngine iterates over all .locked files and decrypts each one using the Fernet cipher. The original file content and filename are fully restored. This strategy models law enforcement key recovery scenarios and demonstrates the cryptographic reversibility of symmetric encryption.

Strategy C — Network Isolation

This strategy simulates the containment phase of incident response. The GUI logs a series of mock firewall and network commands representing: blocking outbound connections to the C2 server IP address, disabling the affected network interface, and alerting the SOC team. While this strategy does not recover files, it prevents further encryption and stops the encryption key from being transmitted to the attacker — a critical containment action in live incidents.

Blue Team Defense Strategies

Characteristic	Action	Effectiveness	Speed	Complexity
Restore from Backup	Restore pre-attack backup copy	High	Moderate	Moderate
Decrypt with Key	Use captured AES key to decrypt	High	Fast	Moderate
Network Isolation	Simulate firewall rules, block C2 traffic, disable SMB	High	Moderate	High

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4.4 Threat Intelligence and Scoring

Throughout the simulation, the ThreatIntelligence module continuously records every event with a millisecond-precision timestamp. After the defense phase, it calculates:

- Time to Detect (TTD): The elapsed time from the first IoC (file encryption start) to the first defensive action
- Files Encrypted vs Files Recovered: A ratio that contributes directly to the Defence Score
- Defence Score (0–100): A composite metric rewarding early detection, complete recovery, and minimal TTD — designed to reflect real-world SOC performance benchmarks

A complete incident_report.json is generated at the end, containing all IoCs, the incident timeline, TTD, and the final Defence Score — suitable for use as a forensic artefact in post-incident review exercises.

5. Overall Functionality

5.1 GUI Panels and Features

- Attack Control Panel: Buttons to initialise the environment, launch the attack, and monitor progress through a real-time progress bar
- File System View: A live-updating list showing the state of all files in the sandbox — normal, .locked (encrypted), or restored — colour-coded for immediate visual comprehension
- IoC and Log Panel: A scrollable timestamped log of every event in the simulation, from initial access through recovery, mirroring the kind of SIEM (Security Information and Event Management) output an analyst would review
- Defense Strategy Panel: Interactive buttons for selecting and executing each of the three recovery strategies
- Analytics Dashboard: Displays TTD, files encrypted, files recovered, and the final Defence Score in a clear, readable format

5.2 Security and Safety Guarantees

The simulation is designed with explicit safety constraints to ensure it is appropriate for academic use:

- All file operations are strictly confined to the craas_test_files/ sandbox directory — no real user files or system directories are ever accessed
- The encryption key exists only in memory during the session and is never written to disk unprotected
- Network isolation strategy is entirely simulated through log output — no actual network interfaces are modified
- All generated files (sandbox, backups, logs, reports) are cleanly removed upon session end via the FileManager cleanup routine

5.3 Educational Outcomes

6. Attacker Perspective: Students gain hands-on understanding of how ransomware encrypts files at the code level, making abstract concepts concrete
7. Defender Perspective: Direct comparison of three recovery strategies builds intuition for why offline backups are the most effective and reliable defence
8. Forensic Analysis: The incident timeline, IoC log, and JSON report closely mirror the artefacts a digital forensics examiner would collect and analyse after a real ransomware incident
9. Performance Measurement: The Defence Score and TTD metric introduce students to the concept of measuring and benchmarking incident response effectiveness

5.4 Limitations and Future Enhancements

Current limitations:

- The simulation models a single-host scenario; lateral movement and network propagation (as seen in worms like WannaCry) are not simulated
- The C2 communication is entirely simulated through log messages; no actual network traffic is generated
- The encryption key is not genuinely exfiltrated, meaning Strategy B always succeeds — in a real attack, the key would be held ransom

Potential future enhancements:

- Integration of a simulated C2 server using a loopback network socket to demonstrate key exfiltration and retrieval
- Multi-host simulation using virtual machines to demonstrate lateral movement and domain-wide encryption
- Integration of a honeypot file detector that triggers an alert at the first sign of mass file modification
- YARA rule-based detection module to identify ransomware behaviour patterns before full encryption occurs

6. Conclusion

This activity successfully demonstrates the full lifecycle of a ransomware attack and the corresponding incident response strategies within a Cyber Range as a Service framework. The GUI-based Python simulation

provides an immersive, safe, and educationally rich environment in which students can experience both the offensive and defensive dimensions of one of today's most consequential cyber threats.

The simulation makes clear that the most effective defence against ransomware is proactive preparation — specifically, maintaining current, isolated, and tested backups — rather than reactive measures taken after encryption has occurred. The Defence Score and Time to Detect metrics reinforce the importance of rapid detection and response in minimising the impact of a ransomware incident.

By simulating the technical mechanisms of AES encryption, kill chain execution, IoC generation, and forensic reporting, this activity equips MSc DFIS students with practical knowledge that directly supports careers in digital forensics, incident response, and security operations.

7. References

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